

Unit 5 Pre-Test

The Pythagorean Theorem

Standards assessed: NC.8.G.6, NC.8.G.7, NC.8.G.8

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.G.7 Apply the Pythagorean theorem to find the hypotenuse.

• I can use $a^2 + b^2 = c^2$ to find the hypotenuse of a right triangle.

Match each vocabulary term to its definition, then find each hypotenuse.

C ▾ hypotenuse ✓	A. The two sides that form the right angle.
A ▾ legs ✓	B. $a^2 + b^2 = c^2$, relating the sides of a right triangle.
B ▾ Pythagorean theorem ✓	C. The side opposite the right angle; the longest side.
D ▾ right triangle ✓	D. A triangle that contains one 90° angle.
E ▾ square root ✓	E. A number that, multiplied by itself, gives a given value.
F ▾ converse ✓	F. If $a^2 + b^2 = c^2$, then the triangle is a right triangle.

Pythagorean theorem — reference

$$a^2 + b^2 = c^2$$

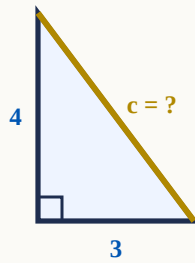
c is the hypotenuse (opposite the right angle); a and b are the legs.

1. Find c

Legs 3 and 4. Find the hypotenuse.

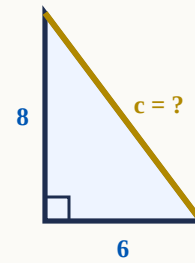
2. Find c

Legs 6 and 8. Find the hypotenuse.



$c =$

answer: 5

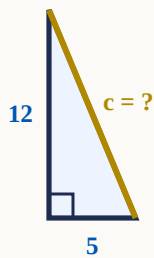


$c =$

answer: 10

3. Find c

Legs 5 and 12. Find the hypotenuse.

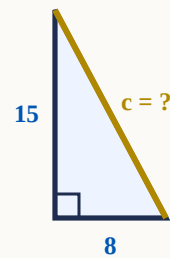


$c =$

answer: 13

4. Find c

Legs 8 and 15. Find the hypotenuse.

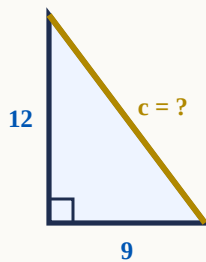


$c =$

answer: 17

5. Find c

Legs 9 and 12. Find the hypotenuse.

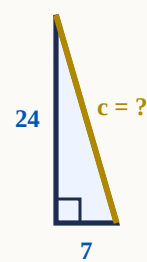


$c =$

answer: 15

6. Find c

Legs 7 and 24. Find the hypotenuse.



$c =$

answer: 25

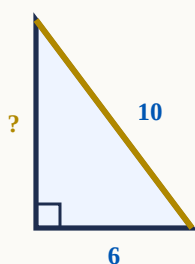
NC.8.G.7 Find a missing leg and estimate non-perfect square roots.

• I can find a missing leg and round answers to the nearest tenth when needed.

Find each missing side. Round to the nearest tenth where asked.

7. Find a leg

One leg 6, hypotenuse 10. Find the other leg.

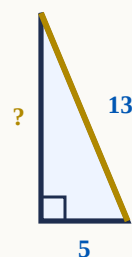


a =

answer: 8

8. Find a leg

One leg 5, hypotenuse 13. Find the other leg.

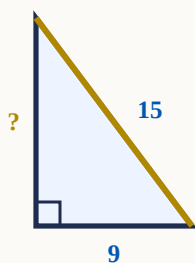


a =

answer: 12

9. Find a leg

One leg 9, hypotenuse 15. Find the other leg.

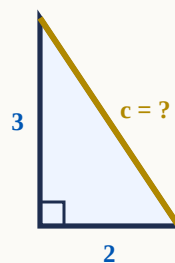


a =

answer: 12

10. Nearest tenth

Legs 2 and 3. Find the hypotenuse to the nearest tenth.



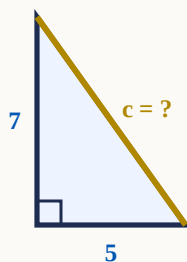
c ≈

answer: 3.6

11. Nearest tenth

12. Nearest tenth

Legs 5 and 7. Find the hypotenuse to the nearest tenth.



$c \approx$

answer: 8.6

One leg 4, hypotenuse 9. Find the other leg to the nearest tenth.



$a \approx$

answer: 8.1

SCORE 5.0 Advanced Content

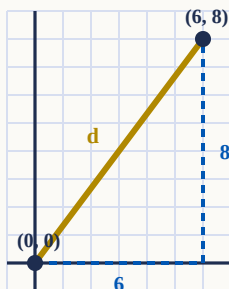
NC.8.G.8 Apply the theorem to the coordinate plane and real-world problems.

• *I can find the distance between two points and solve real-world right-triangle problems.*

Solve each problem.

13. Distance

Find the distance between $(0, 0)$ and $(6, 8)$.

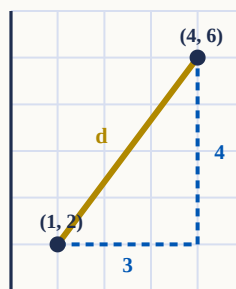


$d =$

answer: 10

14. Distance

Find the distance between $(1, 2)$ and $(4, 6)$.



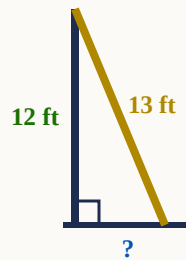
$d =$

answer: 5

15. Apply

16. Apply

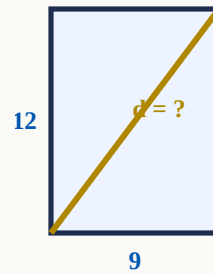
A 13 ft ladder reaches 12 ft up a wall.
How far is the base from the wall?



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answer: 5 ft

A rectangle is 9 by 12. Find the length of its diagonal.

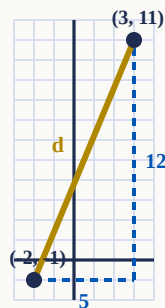


d =

answer: 15

17. Distance

Find the distance between $(-2, -1)$ and $(3, 11)$.

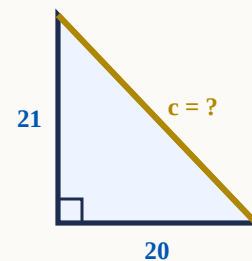


d =

answer: 13

18. Apply

A park is a right triangle with legs 20 m and 21 m. Find the longest side.



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answer: 29 m